

INTERNSHIP REPORT

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Dhaani Bahl

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INTERNSHIP CERTIFICATE

This is to certify that the “INTERNSHIP REPORT” submitted by Dhaani Bahl 22BCE11071 for the internship done by her during the academic year 2025-2026 from 06-02-2025 to 23-05-2025 (3 Months), is the partial fulfillment of the requirements for the awarding the degree of B. Tech – CSE.

Internship In-Charge

Dr. Jayakumar Sadashivam
Assistant Professor

Program Chair

Dr. Dheeresh Soni
Associate Professor

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CERTIFICATION





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I would like to express my heartfelt gratitude to VIT Bhopal University for providing the infrastructure and support that made this internship opportunity possible. I am especially thankful to the Placement Office for their continuous efforts in connecting students with valuable industry-oriented learning programs. Through their initiative, I had the opportunity to pursue the professional courses **DevOps Fundamentals and Agile Methodologies by IBM** and **Machine learning by SmartBridge in collaboration with Google for Developers**, which significantly enhanced my technical understanding and practical exposure to modern technologies. I would also like to sincerely thank IBM and SmartBridge for designing structured learning modules that helped bridge the gap between academic knowledge and real-world industry applications. Special appreciation goes to the Programme Chair and faculty members for offering such relevant courses aligned with current technological trends and for providing constant guidance throughout the duration of the program. Their encouragement and academic support during these intensive months of learning played a crucial role in strengthening my technical and professional skills. This experience has greatly contributed to my growth as a student and has provided valuable insight into practical industry tools and workflows. Finally, I extend my sincere thanks to all mentors and administrative staff whose efforts behind the scenes ensured a smooth and enriching internship experience.

ABSTRACT

This report outlines the comprehensive learning experience and skill development achieved through the successful completion of two professional certification courses: **DevOps Fundamentals and Agile Methodologies** offered by IBM, and **Machine Learning** conducted by SmartBridge in collaboration with Google for Developers. These courses were undertaken with the objective of gaining both theoretical understanding and practical exposure to modern software engineering practices and data-driven technologies that are shaping the current and future landscape of the IT industry.

The **DevOps Fundamentals and Agile Methodologies** course provided an in-depth understanding of the principles and practices that enable organizations to deliver high-quality software efficiently and reliably. It began with an introduction to the traditional software development lifecycle and its limitations, followed by the evolution of DevOps as a solution to bridge the gap between development and operations teams. Key concepts such as Continuous Integration (CI), Continuous Delivery (CD), and Continuous Deployment were explored in detail, emphasizing how automation plays a critical role in reducing manual effort, minimizing errors, and accelerating the development process.

The course also highlighted the importance of collaboration, communication, and shared responsibility among cross-functional teams. Agile methodologies, including Scrum and Kanban, were discussed as frameworks that promote iterative development, flexibility, and rapid feedback. Concepts such as sprint planning, daily stand-ups, backlog management, and retrospectives were introduced to demonstrate how teams can continuously improve their processes and deliver value to stakeholders. Additionally, exposure to tools and practices related to version control, build automation, testing, and monitoring provided a holistic view of the DevOps lifecycle, which includes planning, coding, building, testing, releasing, deploying, operating, and monitoring applications in real-world scenarios.

In parallel, the **Machine Learning** course offered a strong foundation in the concepts and applications of data analysis and intelligent systems. The course began with an introduction to the significance of machine learning in solving real-world problems across domains such as healthcare, finance, marketing, and technology. It covered the fundamental types of machine learning, including supervised learning, unsupervised learning, and an overview of reinforcement learning. Key topics such as data preprocessing, data cleaning, handling missing values, and feature engineering were emphasized as critical steps in preparing data for model training.

Furthermore, the course provided hands-on experience in building, training, and evaluating machine learning models using Python and relevant libraries. Algorithms such as linear regression, logistic regression, decision trees, and clustering techniques were introduced to demonstrate different approaches to prediction and pattern recognition. Model evaluation techniques, including accuracy, precision, recall, and validation methods, were also discussed to ensure the reliability and performance of the developed models.

Overall, the completion of these two courses has contributed to the development of a well-rounded skill set that combines software development efficiency with data-driven intelligence. While the DevOps and Agile training strengthened the understanding of collaborative development, automation, and efficient project management, the Machine Learning course enhanced the ability to analyze data, build predictive models, and derive meaningful insights.

METHODOLOGIES LEARNED

Methodology: DevOps and Agile methodologies

The **DevOps Fundamentals and Agile Methodologies** course by IBM introduced several key methodologies that are widely used in modern software development to improve efficiency, collaboration, and product quality.

One of the primary methodologies learned was **Agile**, which focuses on iterative and incremental development. Instead of building an entire product at once, Agile promotes breaking the project into smaller, manageable units called iterations or sprints. This approach allows teams to adapt quickly to changing requirements, gather continuous feedback, and deliver functional components at regular intervals. Agile emphasizes customer collaboration, flexibility, and continuous improvement over rigid planning.

Within Agile, specific frameworks such as **Scrum** were explored. Scrum provides a structured approach to managing work through defined roles, events, and artifacts. Key roles include the Product Owner, Scrum Master, and Development Team, each responsible for different aspects of the project. Important practices such as sprint planning, daily stand-up meetings, sprint reviews, and retrospectives help ensure transparency, accountability, and continuous progress. This framework enhances team coordination and helps in identifying and resolving issues early.

Another methodology covered was **Kanban**, which focuses on visualizing workflows and optimizing the flow of tasks. Using tools like boards and cards, Kanban helps teams track progress, limit work in progress (WIP), and identify bottlenecks in real time. Unlike Scrum, Kanban is more flexible and does not require fixed-length iterations, making it suitable for continuous delivery environments.

The course also emphasized DevOps methodology, which integrates development and operations to create a seamless software delivery pipeline. DevOps focuses on automation, collaboration, and continuous monitoring throughout the software lifecycle. Practices such as Continuous Integration (CI) and Continuous Delivery (CD) were key highlights, where code changes are automatically built, tested, and prepared for deployment. This reduces manual intervention, minimizes errors, and accelerates the release cycle.

Additionally, the concept of Infrastructure as Code (IaC) was introduced, where infrastructure setup and configuration are managed using code rather than manual processes. This ensures consistency, scalability, and easier maintenance of systems. Continuous Monitoring and Feedback was another important aspect, enabling teams to track application performance and quickly respond to issues.

Methodology: Machine Learning

The **Machine Learning** course conducted by SmartBridge in collaboration with Google for Developers introduced fundamental methodologies used to build data-driven and intelligent systems. The course focused on the complete machine learning pipeline, starting from data collection and preprocessing to model development and evaluation. A strong emphasis was placed on preparing data through techniques such as data cleaning, handling missing values, normalization, and feature selection, as these steps significantly influence model accuracy and performance.

One of the core methodologies covered was supervised learning, where models are trained on labeled datasets to perform tasks such as prediction and classification. Algorithms like linear regression and classification models helped in understanding how relationships between variables are identified and used for decision-making. In addition, unsupervised learning was explored, where models analyze unlabeled data to detect hidden patterns or groupings, such as clustering. These approaches provided insight into different ways of extracting meaningful information from datasets.

Another important methodology introduced was the concept of model training and testing workflows, where datasets are split into training and testing sets to evaluate model performance on unseen data. This ensures that the model generalizes well rather than simply memorizing the training data. The importance of feature engineering was also highlighted, where relevant features are selected or transformed to improve model efficiency and accuracy.

The course further emphasized model evaluation and optimization techniques, including the use of metrics such as accuracy, precision, recall, and validation strategies. Concepts like bias and variance were discussed to understand common challenges such as underfitting and overfitting. Techniques to improve model performance, including tuning parameters and selecting appropriate algorithms, were also introduced.

Additionally, the methodology of iterative improvement in machine learning was highlighted, where models are continuously refined based on performance results and feedback. This process involves experimenting with different algorithms, adjusting parameters, and improving data quality to achieve better outcomes. Overall, these methodologies provided a structured understanding of how machine learning systems are designed, developed, evaluated, and improved to solve real-world problems effectively.

SKILLS ACQUIRED

The completion of the **DevOps Fundamentals and Agile Methodologies** course by IBM and the **Machine Learning** course by SmartBridge in collaboration with Google for Developers contributed significantly to the development of a diverse and industry-relevant skill set. These skills span across software development practices, data analysis, problem-solving, and the ability to build scalable and intelligent systems.

One of the major skills acquired is proficiency in modern software development practices, particularly through the understanding of DevOps principles. This includes knowledge of Continuous Integration and Continuous Delivery (CI/CD), automation of workflows, and the ability to streamline the software development lifecycle. The course enhanced the ability to work with version control systems, manage code efficiently, and understand deployment strategies, which are essential for building reliable and scalable applications. Additionally, the exposure to Agile methodologies improved the ability to work in dynamic environments, adapt to changing requirements, and deliver results in an iterative and efficient manner.

Another key skill developed is collaboration and team-oriented problem solving. **Agile** frameworks such as **Scrum** and **Kanban** emphasize communication, coordination, and shared responsibility among team members. Through these methodologies, the ability to participate in structured workflows, manage tasks effectively, and contribute to team goals was strengthened. Skills such as planning, organizing, and reviewing work through sprints and feedback loops were also enhanced, which are highly valuable in real-world project environments.

In addition, model development and evaluation skills were acquired through hands-on experience with machine learning algorithms. This includes the ability to build predictive models, evaluate their performance using appropriate metrics, and improve them through optimization techniques. Understanding concepts such as bias, variance, and overfitting helped in developing more accurate and reliable models. These skills are essential for designing intelligent systems that can make informed decisions based on data.

Another important area of skill development is technical proficiency in tools and technologies. Both courses provided exposure to programming using **Python** and working with relevant libraries and tools used in machine learning and software development. Familiarity with such tools enhances the ability to implement concepts practically and work on real-world projects efficiently. It also builds confidence in handling complex systems and large datasets.

Furthermore, the courses helped in developing problem-solving and critical thinking abilities. Whether it is optimizing a software pipeline in DevOps or improving the accuracy of a machine learning model, both domains require a structured approach to identifying problems and implementing effective solutions.

Lastly, these courses contributed to the development of adaptability and continuous learning skills. The technology landscape is constantly evolving, and both DevOps and Machine Learning require staying updated with new tools, techniques, and best practices. Through this learning experience, the ability to quickly grasp new concepts and apply them effectively was enhanced.

RESULTS

The outcomes of the **DevOps Fundamentals and Agile Methodologies** course by IBM and the **Machine Learning** course by SmartBridge in collaboration with Google for Developers are clearly demonstrated through the successful development and implementation of practical, real-world projects in both domains. These results reflect not only theoretical understanding but also the ability to apply learned concepts effectively in solving real-world challenges.

As a key outcome of the DevOps course, a fully functional Continuous Integration and Continuous Delivery (CI/CD) pipeline was designed and implemented. The pipeline was structured to automate the entire software development lifecycle, starting from code integration to testing and deployment. Whenever changes were made to the codebase, the pipeline automatically triggered build processes, executed testing procedures, and ensured that only verified and error-free code progressed to the deployment stage. This significantly reduced manual effort, minimized the chances of human error, and improved overall development efficiency.

The implementation of this CI/CD pipeline demonstrated a strong understanding of core DevOps practices such as automation, version control, continuous testing, and monitoring. It also reflected the ability to integrate multiple stages of development into a seamless workflow. The pipeline ensured faster feedback cycles, allowing issues to be identified and resolved at an early stage. Additionally, the project highlighted the importance of collaboration between development and operations, aligning with the fundamental principles of DevOps.

In the Machine Learning domain, a significant outcome was the development of an H1-B visa prediction model, aimed at predicting the likelihood of visa approval based on historical data. The project involved working with structured datasets and applying a complete machine learning workflow. Initial stages included data preprocessing, where missing values were handled, irrelevant features were removed, and important variables were selected to improve model performance.

Following data preparation, multiple machine learning algorithms were explored and implemented to build predictive models. The model was trained using historical visa application data, enabling it to identify patterns and relationships between different factors influencing visa approval decisions. Evaluation techniques were applied to measure model performance using appropriate metrics, ensuring that the predictions were accurate and reliable. The process also involved refining and optimizing the model to achieve better performance and avoid issues such as overfitting.

The H1-B visa prediction model demonstrated the practical application of machine learning concepts in solving real-world problems. It showcased the ability to analyze complex datasets, extract meaningful insights, and develop predictive solutions that can assist in decision-making processes. The project also reflected strong analytical thinking, problem-solving ability, and technical proficiency in implementing machine learning techniques.

SCOPE

The scope of this report is to present the knowledge, methodologies, skills, and practical outcomes achieved through the completion of the **DevOps Fundamentals and Agile Methodologies** course by IBM and the **Machine Learning** course by SmartBridge in collaboration with Google for Developers. It focuses on understanding how modern software development practices and data-driven technologies can be applied to build efficient, scalable, and intelligent systems.

The scope includes the study of DevOps principles such as Continuous Integration and Continuous Delivery (CI/CD), automation of workflows, and the use of Agile methodologies like Scrum and Kanban for effective project management. It also covers the design and implementation of a CI/CD pipeline, demonstrating how development, testing, and deployment processes can be integrated and automated to improve productivity and software quality.

In addition, the scope extends to the field of Machine Learning, including data preprocessing, model development, evaluation, and optimization techniques. It highlights the development of a predictive model for H1-B visa approval, showcasing the application of machine learning algorithms to analyze real-world datasets and generate meaningful predictions. The report also considers the importance of data quality, feature selection, and performance evaluation in building reliable models.

Furthermore, the scope emphasizes the integration of both domains, illustrating how DevOps practices can support the deployment and maintenance of machine learning models in real-world environments. It provides an overview of how these combined technologies contribute to building robust, scalable, and data-driven applications aligned with current industry standards.

However, the scope is limited to the concepts, tools, and implementations covered within the duration of the courses. Advanced topics and large-scale industry implementations are beyond the scope of this report but provide opportunities for future exploration and learning.

WEEKLY PROGRESS

DevOps and Agile methodologies

Week	Topics Covered	Learning	Project Progress (CI/CD Pipeline)
Week 1	Introduction to DevOps & SDLC	Learned basics of DevOps, SDLC models, and need for DevOps	Planned project structure and understood pipeline workflow
Week 2	Version Control	Learned Adroit pro-learn commands, branching, and folder management	Set up folders and saved initial project code
Week 3	Continuous Integration (CI)	Understood CI concepts, automated builds, and testing	Implemented CI process to trigger builds on code commits
Week 4	Continuous Delivery/Deployment (CD)	Learned deployment strategies and automation tools	Added automated deployment stage to pipeline
Week 5	Agile Methodologies (Scrum, Kanban)	Learned sprint planning, stand-ups, and workflow management	Improved pipeline workflow using iterative approach
Week 6	Monitoring & Optimization	Learned monitoring, logging, and performance improvement	Finalized fully functional CI/CD pipeline with automation and testing

Machine Learning

Week	Topics Covered	Learning	Project Progress (CI/CD Pipeline)
Week 1	Introduction to Machine Learning	Learned types of ML, real-world applications	Understood dataset and defined project problem
Week 2	Data Preprocessing	Data cleaning, handling missing values, feature selection	Cleaned dataset and prepared it for modeling
Week 3	Supervised Learning	Learned regression and classification algorithms	Implemented initial models (regression/classification)
Week 4	Unsupervised Learning	Learned clustering and pattern recognition	Explored patterns in visa dataset
Week 5	Model Evaluation	Accuracy, precision, recall, validation techniques	Evaluated model performance and improved accuracy
Week 6	Model Optimization & Finalization	Learned tuning and optimization techniques	Finalized H1-B visa prediction model and tested results

CONCLUSION

In conclusion, the completion of the **DevOps Fundamentals and Agile Methodologies** course by IBM and the **Machine Learning** course by SmartBridge in collaboration with Google for Developers has provided a comprehensive understanding of both modern software development practices and data-driven technologies. These courses have not only strengthened theoretical knowledge but also enhanced practical skills through hands-on implementation of real-world projects.

Through the DevOps course, a clear understanding of automation, collaboration, and efficient software delivery was developed. The successful implementation of a CI/CD pipeline demonstrated the ability to integrate development, testing, and deployment processes into a streamlined workflow. This highlights the importance of speed, reliability, and continuous improvement in today's fast-paced software development environment.

Similarly, the Machine Learning course contributed to building strong analytical and problem-solving skills. The development of the H1-B visa prediction model showcased the practical application of machine learning concepts, including data preprocessing, model building, and performance evaluation. It reflected the ability to work with real datasets and derive meaningful insights to support decision-making.

Overall, the combination of these two domains has created a well-rounded skill set that bridges software engineering and intelligent systems. The knowledge gained from DevOps ensures efficient development and deployment, while machine learning adds the capability to build predictive and data-driven applications. This integrated learning experience prepares for future challenges in the field of information technology and provides a strong foundation for further growth, innovation, and professional development.

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